

AC-03 - Research submission

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Partial research; the original problem remains open.

Authorship is recorded at the author's request. The affiliation was verified against the Simons Foundation staff profile and current CCB directory on 14 September 2026. These sources identify Sidney Holden as a Flatiron Research Fellow in Biological Transport Networks.

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AC-03: Pair-space reduction and cubic obstructions

Status. This report does not determine the exact complex border rank of 3×3 matrix multiplication. Its numerical conclusion remains

$$17 \leq \underline{R}(M_3) \leq 20.$$

The two endpoints are published results, not new bounds. The retained prior package verifies them and provides an exhaustive rank-17 reduction. This report builds on that reduction with an equivalent three-factor target, exact scheme sections, explicit grouped degenerations, and characteristic-zero obstructions to two substantial classes of cubic-order constructions.

The reduced target is a $17 \times 9 \times 9$ tensor F_E with 35 nonzero coordinates. Its border rank is 17 if and only if $\underline{R}(M_3) = 17$. Its cactus rank is at least 18, but this is *not* a border-rank exclusion. An explicit control tensor in the same format has border rank 17 and cactus rank at least 18, illustrating the gap.

For the four-factor target inherited from the preceding report, all three versions obtained by grouping two of its original matrix factors have exact border rank 17. The displayed degenerations do not split into four-factor rank-one terms. On the exclusion side, a 154 350-column exterior-contraction map rules out cubic-order constructions at all 24 normalized root-of-unity base configurations. A separate integral-kernel and quadratic-ideal certificate rules out the grid-plus-barycenter configuration. An independent 24-pattern calculation excludes a restricted monomial Fourier ansatz for the pair plane.

Every restriction in those statements is essential. Arbitrary limiting configurations, nonreduced limiting schemes, and higher-order degenerations have not been excluded. No matching construction with 17, 18, or 19 terms is supplied.

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1 Problem, conventions, and the inherited reduction

The repository asks for the exact border rank over $\mathbb{C}$ of

$$M_3 = \sum_{i,j,k=1}^3 x_{ij} \otimes y_{jk} \otimes z_{ki} \in A \otimes B \otimes C, \quad \dim A = \dim B = \dim C = 9. \quad (1)$$

The third factor uses the trace convention: contraction gives $\text{tr}(XYZ)$. Border rank permits limits of sums of decomposable tensors; it is not the ordinary tensor rank asked for in AC-02. The public AC-03 statement continues to list the exact value as unresolved.¹

The lower bound 17 is due to Conner–Harper–Landsberg. The upper certificate retained here is Smirnov’s length-20 degeneration.² The previous packages reproduce those bounds by exact arithmetic. The upper certificate has constant tensor M_3 , no negative Laurent residual, and error at most $111|t|$ in Frobenius norm for $0 < |t| \leq 1$. The lower certificate eliminates all candidate Borel-fixed pair data at rank 16.

For a projective variety $V \subset \mathbb{P}W$, define

$$\mathcal{G}_r(V) = \overline{\{\text{span}(p_1, \dots, p_r) : p_i \in V, \dim \text{span}(p_1, \dots, p_r) = r\}} \subset \text{Gr}(r, W). \quad (2)$$

All dimensions of linear spaces in this report are vector-space dimensions. Membership in (2) concerns a limit of *entire spans*. It does not require the limiting space to be spanned by a length- r scheme contained in that space.

Put

$$\begin{aligned} A' &= \text{span}(x_{21}, x_{31}), & B' &= \text{span}(y_{21}, y_{22}, y_{31}, y_{32}), & C' &= \text{span}(z_{31}, z_{32}), \\ L &= A' \otimes B' \otimes C', & \mathcal{S} &= \text{span}(M_3) \oplus L, & X &= \text{Seg}(\mathbb{P}A \times \mathbb{P}B \times \mathbb{P}C). \end{aligned} \quad (3)$$

The 16 coordinate tensors of L have disjoint support from M_3 , so $\dim \mathcal{S} = 17$. Let D have basis d_0 and d_{abc} indexed by those 16 coordinate tensors. Define

$$Q = d_0 \otimes M_3 + \sum_{(a,b,c) \in \mathcal{B}_L} d_{abc} \otimes a \otimes b \otimes c. \quad (4)$$

It has 43 nonzero unit coordinates and is concise in D .

1.1 Audited input from the preceding package

The preceding report proves the following computer-assisted statement. Its complete verifier and unchanged archive are included under `prior/`.

Theorem 1.1 (Inherited single-plane reduction). *One has*

$$\underline{R}(M_3) = 17 \iff \mathcal{S} \in \mathcal{G}_{17}(X) \iff \underline{R}_{D,A,B,C}(Q) = 17.$$

Moreover, any Borel-fixed simultaneous span data of a rank-17 degeneration, after a cyclic rotation of A, B, C , have pair spaces

$$E_{AB} = M_3(C^*) \oplus (A' \otimes B'), \quad E_{BC} = M_3(A^*) \oplus (B' \otimes C'). \quad (5)$$

¹OpenProblemsInNLA, AC-03, statement retrieved 14 September 2026. See source(s) [1] in the Sources section.

²Conner–Harper–Landsberg, *New lower bounds for matrix multiplication and the 3×3 determinant*; Smirnov, *The bilinear complexity and practical algorithms for matrix multiplication*. See source(s) [2, 3] in the Sources section.

The classification starts from 343 Borel pair-space charts, including 29 one-parameter charts. The pair tests and square test leave 55 fixed spaces and no parameter families. All $55^3 = 166\,375$ ordered triples are covered. Exactly 87 pass the three-factor test; they form 29 cyclic orbits, and each surviving intersection is 17-dimensional. After cyclic rotation, each is exactly $\mathcal{S}$. The square threshold uses first jets at 17 general points; its modern framework is discussed by Mańdziuk.³

The necessity in Theorem 1.1 is not merely an apolarity assertion. The proof retains actual pair and triple spans in a projective incidence closure, imposes M_3 in the triple span, and applies the Borel fixed-point theorem. Thus the resulting triple space remains in $\mathcal{G}_{17}(X)$. The square and pair conditions are closed necessary conditions on those same data. The complete enumeration forces the triple space to be $\mathcal{S}$.

Conversely, a local basis of a family of 17-dimensional spans converging to $\mathcal{S}$ lifts M_3 to tensors of rank at most 17. The first-factor flattening of Q gives the last equivalence. Cyclic rotation is used to identify equivalent ordered triples, not to require equal pair spaces.

2 An equivalent 35-coordinate tensor

Let

$$E = \text{span} \left\{ s_{ik} = \sum_{j=1}^3 x_{ij} \otimes y_{jk} : 1 \leq i, k \leq 3 \right\} \oplus (A' \otimes B') \subset A \otimes B. \quad (6)$$

Take a 17-dimensional space D_E with basis e_{ik} and e_{ab} indexed by the nine slices and the eight core coordinate tensors. Its inclusion tensor is

$$F_E = \sum_{i,j,k=1}^3 e_{ik} \otimes x_{ij} \otimes y_{jk} + \sum_{a \in \mathcal{B}_{A'}, b \in \mathcal{B}_{B'}} e_{ab} \otimes a \otimes b. \quad (7)$$

The supports of the two sums are disjoint. Thus F_E has 35 nonzero unit coordinates and is concise in D_E .

Theorem 2.1 (Single-pair equivalence). *With $Z = \text{Seg}(\mathbb{P}A \times \mathbb{P}B)$,*

$$\boxed{\underline{R}(M_3) = 17 \iff E \in \mathcal{G}_{17}(Z) \iff \underline{R}(F_E) = 17.}$$

The forward implication uses the audited exhaustive classification in Theorem 1.1; the reverse implication does not require another classification.

Proof. If $\underline{R}(M_3) = 17$, the simultaneous Borel span data have the first pair plane in (5) after cyclic rotation. A pair span remains a limit of 17 rank-one matrix spans, so $E \in \mathcal{G}_{17}(Z)$.

For any r -dimensional matrix space, its inclusion tensor has border rank r if and only if the space belongs to $\mathcal{G}_r$ of the rank-one matrix variety. Indeed, a convergent basis of rank-one matrix spans supplies the first-factor coefficients of a degeneration of the inclusion tensor. Conversely, conciseness in the r -dimensional first factor makes the image of the flattening continuous on a neighborhood of the limiting tensor. This proves the second equivalence.

Finally, apply the linear map $D_E \rightarrow C$ given by $e_{ik} \mapsto z_{ki}$ and $e_{ab} \mapsto 0$ to (7). It gives M_3 . Border rank cannot increase under a linear map, so $\underline{R}(F_E) = 17$ implies $\underline{R}(M_3) \leq 17$. The inherited lower bound gives equality. $\square$

³Mańdziuk, *Border rank lower bounds beyond weak border apolarity*, submitted 10 September 2026. See source(s) [4] in the Sources section.

2.1 Repeated diagonal blocks

Reorder rows of A by the inner index j first, then i , and columns of B by the inner index first, then k . An arbitrary matrix in E becomes

$$\begin{pmatrix} H & P & R \\ 0 & H & 0 \\ 0 & 0 & H \end{pmatrix}, \quad H \in M_3(\mathbb{C}), \quad (8)$$

where P, R are supported on rows 2,3 and columns 1,2. The dimensions are $9 + 4 + 4 = 17$.

Lemma 2.2. *Every matrix in (8) has rank at least $3 \operatorname{rank} H$.*

Proof. If $q = \operatorname{rank} H$, select a nonsingular $q \times q$ submatrix of H . Select the corresponding rows and columns in each of the three diagonal blocks. The resulting $3q \times 3q$ submatrix is upper block triangular with three copies of the same nonsingular submatrix on its diagonal. Its determinant is nonzero. $\square$

Proposition 2.3 (Reduced pair section). *Scheme-theoretically,*

$$Z \cap \mathbb{P}E = \operatorname{Seg}(\mathbb{P}A' \times \mathbb{P}B').$$

In particular, its linear span has dimension eight.

Proof. Set-theoretically, Lemma 2.2 forces $H = 0$ in any rank-one element of E . The resulting rank-one elements are exactly the small Segre variety.

At a nonzero rank-one matrix $a \otimes b$, the affine tangent space of the rank-one variety is $a \otimes B + A \otimes b$. Every matrix in it has rank at most two. Its intersection with E therefore again has $H = 0$. Its intersection with $A' \otimes B'$ is precisely $a \otimes B' + A' \otimes b$, the tangent space of the small Segre variety. Thus every projective local tangent dimension is four, equal to the local dimension of the set-theoretic intersection. Each local ring is regular. The section is therefore reduced and has exactly the asserted scheme structure. $\square$

2.2 A cactus obstruction, not a border obstruction

The cactus rank $\operatorname{cR}(T)$ is the least length of a zero-dimensional scheme in the appropriate Segre variety whose linear span contains T . This differs from a limit of spans: a flat point-scheme limit can lose span dimension. The distinction is central in border apolarity.⁴

Corollary 2.4. *The tensor F_E satisfies $\operatorname{cR}(F_E) \geq 18$.*

Proof. Suppose a scheme of length at most 17 in $\operatorname{Seg}(\mathbb{P}D_E \times \mathbb{P}A \times \mathbb{P}B)$ spans F_E . Its scheme-theoretic image in Z has length at most 17 and its span contains the flattening image E . A scheme of length at most 17 spans a vector space of dimension at most 17. Since $\dim E = 17$, this image would have to span exactly E and hence be contained in $Z \cap \mathbb{P}E$. Proposition 2.3 places its span in the eight-dimensional core, a contradiction. $\square$

Corollary 2.4 cannot be substituted for $\underline{\operatorname{R}}(F_E) \geq 18$. Section 4.3 gives a verified counterexample to that inference in exactly the same tensor format.

⁴Buczyńska–Buczyński, *Apolarity, border rank and multigraded Hilbert scheme*. See source(s) [5] in the Sources section.

3 The full scheme section of the 17-plane

Let $Y = \text{Seg}(\mathbb{P}A' \times \mathbb{P}B' \times \mathbb{P}C')$, so $\text{span } Y = \mathbb{P}L$ and $\dim Y = 5$. Use coordinates $\lambda, z_1, \dots, z_{16}$ on $\mathcal{S}$, with λ the coefficient of M_3 . Write J_Y for the ideal of the small Segre variety in $\mathbb{C}[z_1, \dots, z_{16}]$.

Theorem 3.1 (Restricted ideal and saturation). *The restriction of the homogeneous Segre ideal to $\mathbb{P}\mathcal{S}$ is*

$$I_X|_{\mathcal{S}} = J_Y + \lambda(\lambda, z_1, \dots, z_{16}). \quad (9)$$

Its saturation is $J_Y + (\lambda)$. Consequently

$$X \cap \mathbb{P}\mathcal{S} = Y$$

scheme-theoretically, with Y reduced.

Proof. The Segre ideal is generated by the 2×2 minors of its flattenings. Restrict a tensor coordinate to $\mathcal{S}$. It is either zero, λ , or one of the 16 distinct variables z_i , because the two support sets are disjoint. The restriction of any flattening minor, after setting $\lambda = 0$, is a small-Segre minor or zero. Thus every restricted minor belongs to the right side of (9).

Conversely, all small-Segre minors are restrictions of large-Segre minors. Here are explicit certificates for the extra 17 monomials, in zero-based row-major coordinate indices. The tensor coordinates of M_3 are $(3i + j, 3j + k, 3k + i)$. In the A -flattening, use diagonal coordinates $(0, 0, 0)$ and $(4, 3, 1)$. The off-diagonal product vanishes, so this minor is λ^2 . For every core coordinate β , use diagonal coordinates $(0, 0, 0)$ and β . The off-diagonal product again vanishes, giving λz_β . These certificates prove equality.

The product of λ with every degree-one coordinate belongs to the ideal, so λ lies in its saturation. The quotient after adding λ is the homogeneous coordinate ring of the reduced Segre variety Y , whose ideal is saturated. This proves the saturation formula. $\square$

The exact quadratic-space dimensions are

$$\dim(J_Y)_2 = 46, \quad \dim(I_X|_{\mathcal{S}})_2 = 63 = 46 + 17.$$

The unsaturated coordinate ring has a one-dimensional extra degree-one class annihilated by the homogeneous maximal ideal. Its diagonal Hilbert function starts

$$1, 17, 90, 320, 875, 2016, 4116, 7680, 13365.$$

For $d \geq 2$ the value is $(d+1)^2 \binom{d+3}{3}$, the Hilbert function of Y . The extra class is at the vertex of the affine cone; it is not a nonreduced projective component of $X \cap \mathbb{P}\mathcal{S}$.

Corollary 3.2. *Any flat limit of the 17 point schemes accompanying a span limit to $\mathcal{S}$ is contained scheme-theoretically in Y . No length-17 scheme in X spans $\mathcal{S}$. The four-factor tensor Q has cactus rank at least 18.*

Proof. Containment of a scheme in its limiting linear span is closed, so a limiting scheme is contained in $X \cap \mathbb{P}\mathcal{S} = Y$. Every scheme in Y spans a subspace of L , whose dimension is 16, not 17. For Q , project a hypothetical length-17 spanning scheme to the last three factors and use first-factor conciseness, exactly as in Corollary 2.4. $\square$

3.1 All positive prolongations

For $a, b, c \geq 1$, let K_{abc} be the space of partially symmetric tensors of multidegree (a, b, c) all of whose contractions to degree $(1, 1, 1)$ belong to $\mathcal{S}$. Equivalently, it is the annihilator in that degree of the ideal generated by $\mathcal{S}^\perp$ in degree $(1, 1, 1)$.

Proposition 3.3. *The first positive prolongations are*

$$\begin{aligned} K_{211} &= \text{Sym}^2 A' \otimes B' \otimes C', & \dim K_{211} &= 24, \\ K_{121} &= A' \otimes \text{Sym}^2 B' \otimes C', & \dim K_{121} &= 40, \\ K_{112} &= A' \otimes B' \otimes \text{Sym}^2 C', & \dim K_{112} &= 24. \end{aligned}$$

More generally, for $a + b + c \geq 4$,

$$K_{abc} = \text{Sym}^a A' \otimes \text{Sym}^b B' \otimes \text{Sym}^c C'.$$

Proof. An element of $A \otimes \mathcal{S}$ has a unique expression $u \otimes M_3 + H$ with $H \in A \otimes L$. Project the B factor to B/B' . The second summand disappears. If the two A factors are symmetric, then $u \otimes \overline{M}_3$ is symmetric in them. If $u \neq 0$, this forces the A -flattening image of $\overline{M}_3$ to lie in $\text{span}(u)$: contract the other factors and use $u \otimes v = v \otimes u$ only when v is proportional to u . But that flattening has rank nine. Each of the nine slices has a nonzero coordinate, and their supports are disjoint. Thus $u = 0$.

Symmetry of $H \in A \otimes L$ then forces the extra A factor into A' . This proves the first formula. The other two follow by the same argument, using a quotient in one of the other factors which preserves at least two independent slices (in fact nine). The supplied wedge-kernel computations check all three statements over $\mathbb{Q}$; their ranks are 129, 113, 129 on 153-dimensional domains.

For higher degree, contract all but one additional slot. The degree-four statements show that every resulting coefficient is supported on the small factor spaces. Repeating this for each slot puts every factor in the corresponding small space. The reverse inclusion is immediate. $\square$

These formulas describe necessary prolongation spaces. They do not force a limiting 17-dimensional degree- $(1, 1, 1)$ span to equal the span of the saturated limiting point scheme.

4 Exact rank-17 certificates after grouping factors

The next construction is a positive result for *grouped* tensors. It also gives an explicit control against an invalid cactus-to-border inference.

4.1 A Fourier compression formula

Let p, q, s be positive integers, $N = pq + 1$, and $p \leq s \leq N - q$. Consider independent basis vectors d_0, d_{rj} in an N -dimensional space, e_k, e'_r in an $(s + q)$ -dimensional space, and c_k in an s -dimensional space. Define

$$T_{p,q,s} = d_0 \otimes \sum_{k=1}^s e_k \otimes c_k + \sum_{r=1}^q \sum_{j=1}^p d_{rj} \otimes e'_r \otimes c_j. \quad (10)$$

Theorem 4.1. *The tensor in (10) has border rank exactly N .*

Proof. Work with residue labels modulo N . Set

$$J = \{0, \dots, p-1\}, \quad R = \{p, 2p, \dots, qp\}.$$

Choose K of size s containing J and disjoint from R , and put $H = K \cup R$. Define $w_j = 0$ on J , $w_h = 2$ on R , and $w_h = 1$ on $K \setminus J$. Use the residue labels to index the corresponding e and c basis vectors. The differences $r - j$ for $r \in R, j \in J$ are exactly the distinct integers $1, \dots, N-1$, so relabel d_{rj} by d_{r-j} .

For a primitive N th root of unity ζ , put

$$\frac{1}{N} \sum_{a=0}^{N-1} \left(d_0 + t^2 \sum_{\ell=1}^{N-1} \zeta^{a\ell} d_\ell \right) \otimes \left(\sum_{h \in H} t^{-w_h} \zeta^{-ah} e_h \right) \otimes \left(\sum_{k \in K} t^{w_k} \zeta^{ak} c_k \right). \quad (11)$$

The root average is zero unless $\ell - h + k = 0$ modulo N , when it is one. For $\ell = 0$, only $h = k$ occurs, with power zero, giving the first summand of (10). For $\ell \neq 0$, the power is $2 + w_k - w_h \geq 0$. It is zero exactly when $h \in R, k \in J$, giving all the second summands. All remaining terms have positive powers. Thus (11) is an N -term degeneration of (10).

The flattening in the d factor has rank N : the core matrix units are independent, and the diagonal $e_k \otimes c_k$ sum has disjoint support. This gives the matching lower bound. $\square$

The verifier works at $N = 17$ and checks root orthogonality in $\mathbb{Q}[\zeta]/(1 + \zeta + \dots + \zeta^{16})$. It expands the complete coefficient tensor, not just its prescribed nonzero entries.

4.2 Application to the four-factor target

In Q , keep D separate and keep one of A, B, C as a singleton. Group the other two matrix factors. The nine slices of M_3 in the grouped factor are independent and disjoint in support from its q core coordinate vectors. Thus the grouped tensor is linearly equivalent to (10):

Singleton	p	q	s	Grouped factor dimension	N
A	2	8	9	17	17
B	4	4	9	13	17
C	2	8	9	17	17

Corollary 4.2. *The three grouped border ranks of Q just described are exactly 17.*

The coefficient expansions have residual degrees 1, 2, 3. Their total coordinate ℓ^1 residual coefficients are 272, 236, 272, respectively, so the same numbers times $|t|$ are Frobenius error bounds for $0 < |t| \leq 1$. At $t = 1$ and Fourier index zero, each grouped middle factor, reshaped in the two original factors, has matrix rank four. These particular middle factors are not decomposable. The formula is therefore not a four-factor border-rank-17 decomposition of Q .

4.3 A control tensor with border rank below cactus rank

Let $a_0, \dots, a_8$ and $b_0, \dots, b_8$ be bases and set

$$W = d_0 \otimes \sum_{i=0}^8 a_i \otimes b_i + \sum_{i=0}^7 \sum_{j=0}^1 d_{1+2i+j} \otimes a_i \otimes b_j \in \mathbb{C}^{17} \otimes \mathbb{C}^9 \otimes \mathbb{C}^9. \quad (12)$$

This is a different tensor from F_E .

Theorem 4.3 (Exact counterexample to a tempting inference). *The tensor W has $\underline{R}(W) = 17$ and $\text{cR}(W) \geq 18$.*

Proof. Apply Theorem 4.1 at $(p, q, s) = (2, 8, 9)$, sending the nine e_k to $a_0, \dots, a_8$ and the eight e'_r to $a_0, \dots, a_7$. This linear restriction gives (12), so its border rank is at most 17. The D -flattening has rank 17, giving equality. The supplied expansion has 25 constant unit coordinates and residual coordinate norms 70, 44, 14 in degrees 1, 2, 3.

The matrix plane of W is $E_W = \text{span}(I_9) \oplus (\text{span}(a_0, \dots, a_7) \otimes \text{span}(b_0, b_1))$. A matrix with nonzero I_9 coefficient has rank at least seven: its columns 2 through 8 are independent scalar multiples of the corresponding standard basis vectors. Hence its rank-one locus is the core Segre variety, whose span has dimension 16. The same rank-at-least-seven versus tangent-rank-at-most-two argument as in Proposition 2.3 proves that the section is reduced. First-factor conciseness then excludes a spanning scheme of length 17, as before. $\square$

This control is included to prevent a false completion of AC-03 from Corollary 2.4. It is not used to assert that F_E has border rank 17. No claim of publication-level novelty is made for this compression family or control; related constructive minimal-border-rank and wild-tensor phenomena occur in the recent literature.⁵

5 Necessary conditions for cubic-order constructions

Suppose 17 normalized holomorphic factor curves have limiting points in Y :

$$a_s(t) \otimes b_s(t) \otimes c_s(t), \quad 1 \leq s \leq 17.$$

Normalization means that each factor curve has a nonzero limit. Scalar coefficients are allowed poles. Their order is measured *after* this normalization; shifting powers between a factor and its scalar would otherwise make a pole-order statement meaningless.

Let π project all three factors to $A/A', B/B', C/C'$. Then

$$\begin{aligned} \pi(M_3) = & \bar{x}_{11} \otimes \bar{y}_{11} \otimes \bar{z}_{11} + \bar{x}_{11} \otimes \bar{y}_{12} \otimes \bar{z}_{21} \\ & + \bar{x}_{32} \otimes \bar{y}_{23} \otimes \bar{z}_{33} + \bar{x}_{33} \otimes \bar{y}_{33} \otimes \bar{z}_{33}. \end{aligned} \quad (13)$$

Its three flattening ranks are $(3, 4, 3)$.

Lemma 5.1. *No such expression for M_3 can have all scalar poles of order at most two.*

⁵Kassabov–Landsberg–Souza–Speegle, *Disjoint and nearly disjoint sums of matrix multiplication tensors and their centroids*. See source(s) [7] in the Sources section.

Proof. Each projected factor is $O(t)$, so every projected decomposable term is $O(t^3)$. Poles of order at most two leave a sum tending to zero, contrary to (13). $\square$

For the cubic test, assume the 17 base points have 16 independent triple products with a unique relation whose 17 coefficients are nonzero. Identify functions on the 17 points with the semisimple algebra $R = \mathbb{C}^{17}$. Let $A_0, B_0, C_0 \subset R$ be the factor-coordinate function spaces, of dimensions 2, 4, 2, and write the relation as a functional $\lambda : R \rightarrow \mathbb{C}$ with

$$\lambda(A_0 B_0 C_0) = 0.$$

The bilinear form $(f, g) \mapsto \lambda(fg)$ is nondegenerate because all relation coefficients are nonzero.

The first normal function spaces are

$$U_A = \text{ann}(B_0 C_0)/A_0, \quad U_B = \text{ann}(A_0 C_0)/B_0, \quad U_C = \text{ann}(A_0 B_0)/C_0, \quad (14)$$

where annihilators use this bilinear form. When the pair product dimensions are 8, 4, 8, these spaces have dimensions 7, 9, 7.

Lemma 5.2 (Second-normal conditions). *A cubic-pole expression for M_3 at these base points supplies subspaces of U_A, U_B, U_C of dimensions at least (3, 4, 3) satisfying*

$$\lambda(ABC_0) = 0, \quad \lambda(AB_0C) = 0, \quad \lambda(A_0BC) = 0. \quad (15)$$

The all-normal constant tensor is the restriction of $(f, g, h) \mapsto \lambda(fgh)$ to the first normal images. It must equal (13) after the physical coordinate maps.

Proof. The coefficient of t^{-3} is a relation among the base products, hence a scalar multiple of the unique λ . If it were zero, Lemma 5.1 would apply. Project one factor outside its small space. The coefficient of t^{-2} then says that each first normal coordinate annihilates the product of the other two base coordinate spaces. This gives (14).

Project two factors outside. The coefficient of t^{-1} involves exactly their first normal coordinates and the base value of the third factor, which gives (15). Project all three factors: the constant term contains exactly three first normal coordinates and the leading scalar relation. Its flattening ranks must be those of (13).

Passing to the quotient representatives in (14) is legitimate. Terms added from a base space vanish in the second-normal conditions by the first conditions; they vanish in the cubic tensor by (15). Thus all the stated restrictions are well defined. $\square$

5.1 An exterior-contraction certificate

For a bilinear form m on $U \times V$, define

$$\partial_m(u_1 \wedge \cdots \wedge u_a \otimes v_1 \wedge \cdots \wedge v_b) = \sum_{i,j} (-1)^{i+j} m(u_i, v_j) u_1 \wedge \cdots \widehat{u}_i \cdots \wedge u_a \otimes v_1 \wedge \cdots \widehat{v}_j \cdots \wedge v_b.$$

Numbering signs from zero instead of one gives the same formula. Apply all the forms in (15) to

$$\mathcal{V} = \Lambda^3 U_A \otimes \Lambda^4 U_B \otimes \Lambda^3 U_C, \quad \dim \mathcal{V} = \binom{7}{3} \binom{9}{4} \binom{7}{3} = 154\,350. \quad (16)$$

Let $\mathcal{D}$ be their joint contraction map.

Lemma 5.3. *If $\mathcal{D}$ is injective, no subspaces of dimensions $(3, 4, 3)$ satisfy (15). Therefore no cubic-pole expression for M_3 exists at the specified base configuration.*

Proof. Bases of such subspaces give a nonzero decomposable product of three Plücker vectors in (16). Every contraction is zero by (15), contradicting injectivity. $\square$

6 All normalized cyclic-root base configurations

Take $R = \mathbb{C}[z]/(z^{17} - 1)$ and choose exponent sets A_0, B_0, C_0 of sizes 2, 4, 2 whose triple sums are 16 distinct residues. The 17 evaluation points at the roots of unity give 16 independent base products. If m is the missing residue, the unique relation is $\lambda(f) = [z^m]f$. Its evaluation weights are nonzero. Products and pairings have rational structure constants in the residue basis.

Translations of the three exponent sets, interchange of their basis labels, and rescaling $z \mapsto z^a$, $a \not\equiv 0 \pmod{17}$, normalize the sets to

$$A_0 = \{0, 1\}, \quad B_0 = \{0, b_1, b_2, b_3\}, \quad C_0 = \{0, c\}, \quad 1 \leq b_1 < b_2 < b_3 \leq 16. \quad (17)$$

Exactly 24 normalized configurations have 16 distinct triple sums. This is a finite enumeration of *exponent configurations*, not an enumeration of arbitrary complex point configurations.

Theorem 6.1 (Cyclic cubic obstruction). *For every configuration in (17) with 16 distinct triple sums, the map $\mathcal{D}$ in (16) has full column rank 154 350 over $\mathbb{Q}$ and $\mathbb{C}$. Hence no normalized holomorphic factor curves with those limiting points and scalar poles of order at most three can express M_3 in the limit.*

Computer-assisted proof. For each configuration, the normal quotient spaces are obtained exactly from the rule $\lambda(z^a z^b) = 1$ when $a + b = m$ modulo 17 and zero otherwise. Their dimensions are 7, 9, 7. The coefficient of a contraction column is 0, 1, or -1 .

First apply the two AB forms to $\Lambda^3 U_A \otimes \Lambda^4 U_B$, of dimension 4410. The computation splits by the sum of character labels modulo 17. It constructs a kernel basis modulo 101 and independently checks every kernel relation against the original columns. Tensor that kernel with $\Lambda^3 U_C$, then apply the remaining four AC and two BC forms. Each residual character block has full column rank. The combined rank is exactly 154 350 in each of the 24 cases.

A full-column-rank reduction modulo a prime certifies a nonzero maximal minor of the original integer matrix. It therefore certifies injectivity in characteristic zero. There is no claim that testing complex subspaces can be replaced by enumerating subspaces over a finite field. Lemma 5.3 completes the proof. $\square$

The two configurations

$$(\{0, 1\}, \{0, 2, 4, 6\}, \{0, 8\}), \quad (\{0, 1\}, \{0, 2, 8, 10\}, \{0, 4\})$$

are independently checked modulo 103 as well. Appendix A also gives a hand proof for the first, independent of the exterior rank calculation.

Corollary 6.2 (A generic reduced-configuration obstruction). *There is a nonempty Zariski-open subset of the ordered 17-point configurations in Y for which cubic-pole expressions for M_3 are impossible.*

Proof. Restrict to the nonempty open set where triple evaluation has rank 16, the unique relation has no zero coefficient, and the pair product spaces have maximal dimensions. The normal spaces in (14) form vector bundles of ranks 7, 9, 7. On local charts their contractions depend algebraically on the points. Injectivity of $\mathcal{D}$ is open. The first configuration in Theorem 6.1 proves this open set is nonempty. The configuration variety is irreducible, so the open subset is also dense. $\square$

The corollary does not cover the exceptional closed locus or higher poles. In particular, the grid in the next section has a nonzero exterior kernel, so a blanket injectivity statement would be false.

7 The grid-plus-barycenter obstruction

Take the 16 coordinate product points of Y and the product of the three barycenter vectors. Normalize each barycenter factor so its coordinates sum to one. The product at the barycenter is $1/16$ times the sum of the 16 grid products, so the unique relation has weights -1 on the grid and 16 on the barycenter.

Use the group $G = (\mathbb{Z}/2)^4$ to label functions on the grid. The constant function is one on all 17 points; each nontrivial character is extended by zero at the barycenter. A Fourier change of basis within the factors gives base character spaces

$$A_0 = \langle 1, \chi_1 \rangle, \quad B_0 = \langle 1, \chi_2, \chi_4, \chi_6 \rangle, \quad C_0 = \langle 1, \chi_8 \rangle. \quad (18)$$

For nontrivial characters, a relation moment is -16 when the binary exclusive-or sum of the labels is zero and otherwise zero. The irrelevant common factor -16 is dropped in the matrices.

The three normal quotient bases are

$$\begin{aligned} U_A &= (3, 5, 7, 9, 11, 13, 15), \\ U_B &= (3, 5, 7, 10, 11, 12, 13, 14, 15), \\ U_C &= (9, 10, 11, 12, 13, 14, 15). \end{aligned}$$

The joint exterior map is *not* injective here. Its kernel has dimension 135, and genuine $(3, 4, 3)$ subspaces satisfying the pair conditions exist. Appendix B displays one. Further equations are necessary.

Theorem 7.1 (Grid cubic obstruction). *For the configuration (18), every triple of normal subspaces of dimensions $(3, 4, 3)$ satisfying (15) has identically zero cubic form $\lambda(fgh)$ on their product. Consequently the grid-plus- barycenter configuration cannot be the base of a cubic-pole expression for M_3 .*

Exact certificate and its algebraic interpretation. Let $\mathcal{K} = \ker \mathcal{D} \subset \mathcal{V}$. The proof uses five finite steps; all polynomial operations after the initial modular rank bound are over $\mathbb{Q}$.

The complete linear kernel. The file `grid_joint_kernel.json` gives 135 integral vectors in $\mathcal{V}$. Each is substituted into the original contraction columns, and its image is checked to be zero over $\mathbb{Q}$. Their independence is checked by rational elimination. Independently, a block computation modulo 101 gives rank $154\,215 = 154\,350 - 135$ for the original integer map. A characteristic-zero kernel has dimension at most the modular kernel dimension. The 135 independent rational vectors therefore form the entire kernel over $\mathbb{Q}$ and, by scalar extension, over $\mathbb{C}$.

Necessary quadratic equations. A candidate product of Plücker vectors must be decomposable across the three exterior factors and must have a Grassmannian Plücker vector in each factor. Thus its coordinates satisfy the three-factor Segre minors and the Grassmann Plücker relations on every fixed fiber. Substitute a general linear combination of the 135 kernel vectors into these equations. Two batches of explicit Segre minors reduce to nonzero scalar multiples of individual parameter squares and eliminate 87 parameters, leaving 48. The selected, explicitly recorded necessary quadrics span a 1143-dimensional subspace of the 1176-dimensional space of quadrics in those 48 parameters. It is not necessary to assert that these selected quadrics generate the whole ideal.

Linear consequences of quadratic ideal membership. There are 17 independent linear forms ℓ whose products ℓx_j with every remaining parameter lie in the selected quadratic span. The verifier computes them as a rational kernel and checks those memberships. Therefore ℓ^2 belongs to that span, and every complex solution has $\ell = 0$. Restricting to their common zero space leaves 31 parameters. The restricted quadratic span has rank 463.

The cubic contraction. Contract one exterior slot in each factor against $\tau(f, g, h) = \lambda(fgh)$. This defines

$$\partial_\tau : \mathcal{V} \longrightarrow \Lambda^2 U_A \otimes \Lambda^3 U_B \otimes \Lambda^2 U_C.$$

On the remaining 31-parameter linear space its coefficient functionals span a seven-dimensional space. Three independent such functionals have squares in the restricted necessary quadratic span; hence they vanish at every complex solution. After imposing them, 28 parameters remain and the image of ∂_τ lies in a four-dimensional tensor space.

An exact flattening contradiction. For a basis $V_1, \dots, V_4$ of that four-dimensional image, each V_i has 23 columns in its first-factor flattening which occur in no other V_j . Those exclusive columns have rank nine. Therefore every nonzero linear combination of the V_i has first flattening rank at least nine: choose a nonzero coefficient and restrict to its exclusive columns.

On the other hand, if the original exterior vector is a product of Plücker vectors of spaces A, B, C of dimensions 3, 4, 3, its cubic contraction lies in

$$\Lambda^2 A \otimes \Lambda^3 B \otimes \Lambda^2 C.$$

Its first flattening rank is at most $\dim \Lambda^2 A = 3$. It must therefore be zero. The cofactor wedge bases in these three spaces are independent, so this is equivalent to τ being zero on $A \times B \times C$.

If larger first normal image spaces existed for M_3 , any nonzero cubic coordinate could be extended to subspaces of dimensions (3, 4, 3), since the projected flattening ranks are (3, 4, 3). The preceding conclusion would make that coordinate zero. Thus Lemma 5.2 contradicts (13) in every case. $\square$

This proof neither infers complex nonexistence from unsuccessful numerical searches nor treats a modular kernel as an exact rational kernel without checking it. The integral basis plus the one-sided modular rank bound provides the required characteristic-zero equality. The additional quadratic equations are essential: the pair conditions alone admit the control in Appendix B.

8 A separate restricted monomial Fourier exclusion

There is a different root-of-unity search for the pair plane E , distinct from the cubic-base tests above. Consider curves indexed by $s \in \mathbb{Z}/17$ whose every matrix coordinate has one nonzero character monomial:

$$a_{ij}^{(s)}(t) = c_{ij} t^{\alpha_{ij}} \zeta^{se_{ij}}, \quad b_{\ell k}^{(s)}(t) = d_{\ell k} t^{\beta_{\ell k}} \zeta^{sf_{\ell k}},$$

with all $c_{ij}, d_{\ell k} \neq 0$. Arbitrary Laurent or rational exponents may be allowed; the obstruction below already holds for real exponents.

A span limit equal to E forces the nine repeated diagonal slices and the eight core units to occupy 17 distinct characters. Equality of characters within a diagonal slice implies

$$e_{ij} = u_i + v_j, \quad f_{jk} = w_k - v_j \pmod{17}.$$

Equality of valuations within each such slice likewise gives $\alpha_{ij} = a_i + b_j$ and $\beta_{jk} = c_k - b_j$. Normalize $u_0 = w_0 = v_0 = 0$, $u_1 = 1$, and interchange the last two inner labels so $v_1 < v_2$. There are 810 possible distinct nine-character main sumsets; exactly 24 character patterns also assign eight disjoint characters to the core units.

Every off-target matrix coordinate in a given character must have strictly larger valuation than that character's prescribed limiting slice or unit. There are 46 such strict inequalities in six independent real weights.

Theorem 8.1 (Restricted monomial obstruction). *No curves of the preceding single-character, single-monomial form have 17-dimensional pair spans tending to E .*

Exact finite certificate. The verifier enumerates all 810 main sumsets and all 24 normalized full patterns. For each full pattern it constructs the 46 integer gap rows $g_i \in \mathbb{Z}^6$. The certificate supplies positive rational coefficients q_i , on a nonempty subset of these rows, satisfying

$$\sum_i q_i = 1, \quad \sum_i q_i g_i = 0.$$

If all required gaps $g_i \cdot w$ were positive, their positive weighted sum would be positive, contradicting the displayed zero identity. Every row and rational dependence is regenerated and checked. Thus all 24 patterns are impossible, even over arbitrary real valuation weights. $\square$

Nonzero amplitudes cannot cancel an off-target coordinate in a fixed character column: it is a single product of two nonzero amplitudes. This justifies the strict-gap necessity. The theorem does not cover sums of character monomials in a coordinate, arbitrary factor curves, or non-diagonal changes of the ambient coordinates during the degeneration. The coincidence that this enumeration and Section 6 each contain 24 normalized patterns does not make them the same classification.

9 What remains to determine the exact border rank

The new pair reduction sharpens the unresolved rank-17 question to

$$E \stackrel{?}{\in} \mathcal{G}_{17}(\text{Seg}(\mathbb{P}A \times \mathbb{P}B)), \quad \text{equivalently} \quad \underline{R}(F_E) \stackrel{?}{=} 17. \quad (19)$$

A positive certificate must provide a genuine limiting span or an exact 17-term degeneration of F_E . A universal nonmembership proof would establish $\underline{R}(M_3) \geq 18$. It would *not* by itself determine whether the answer is 18, 19, or 20.

The results here eliminate a monomial Fourier route and cubic constructions at the specified cyclic and grid bases. They do not eliminate arbitrary special reduced configurations, nonreduced limits, or higher normalized scalar poles. Corollaries about cactus rank do not fill that gap, as the control

tensor proves. The positive grouped certificates do not fill it either, because grouping forgets the required decomposability of a merged factor.

A separate exploratory route considers unrestrictions to concise minimal-border-rank tensors and necessary commutative-algebra conditions. The general framework is given by Jagiełła–Jelisiejew.⁶ The exploratory basis-line enumeration in this package is incomplete at parameter-bearing branches and is not used for an exclusion. Finite-field samples and floating-point optimization records are likewise not proofs of nonexistence over $\mathbb{C}$.

The exact numerical interval is unchanged. None of the arguments in this report selects a value from 17, 18, 19, 20, and the package must not be represented as a completed solution of AC-03.

10 Reproducibility and audit boundaries

From the extracted directory, run

```
python code/check_manifest.py
python -m pip install -r requirements.txt
python code/verify_all.py
```

The new certificates use only Python’s standard library. SymPy 1.14.0 is needed for the retained prior verifier. The tested interpreter is Python 3.13.5. Do not use Python’s `-O` option: assertions are verification checks, and the combined runner rejects optimized mode.

The combined runner regenerates the scheme-section, pair-plane, grouped, control, cyclic, monomial, exterior, and grid results. It runs eight consistency tests and independently checks the two principal cyclic configurations modulo 103. It extracts the unchanged previous archive into a temporary directory, verifies its manifest, and recomputes its complete round-two and round-one checks. The optional `-skip-prior` switch checks only the new work and must not be described as revalidating the inherited bounds or classification.

File	Role
<code>code/pair_geometry.py</code>	35-coordinate target, restriction, block conventions.
<code>code/section_geometry.py</code>	Restricted ideal and complete first prolongation kernels.
<code>code/grouped_degeneration.py</code>	All three exact grouped length-17 expansions.
<code>code/wild_control.py</code>	Independent border-17/cactus-at-least-18 control.
<code>code/exterior_obstruction.py</code>	Complete cyclic-root normalization and integer-rank certificates.
<code>code/grid_obstruction.py</code>	Integral kernel, necessary quadrics, and cubic flattening obstruction.
<code>code/toric17_verify.py</code>	All monomial patterns and positive rational gap certificates.
<code>certificates/</code>	Exact input witnesses, not expected numerical answers alone.
<code>results/</code>	Regenerated proof records and verification logs.
<code>research/</code>	Exploratory calculations, with incompleteness and sampling boundaries.

⁶Jagiełła–Jelisiejew, *Unrestrictions and concise secant varieties*, version 2. See source(s) [6] in the Sources section.

The report contains the logical reductions interpreting the finite calculations. Universal rank and scheme arguments are not inferred from sample matrices. The modular computations prove nonzero integer minors or bound a rational kernel which is independently constructed; they do not enumerate complex solutions. The checks are computer-assisted mathematics, not proof-assistant formalization, independent peer review, or a guarantee against every implementation error.

A A hand proof for one cyclic configuration

This appendix proves the cubic obstruction for

$$A_0 = \langle 1, z \rangle, \quad B_0 = \langle 1, z^2, z^4, z^6 \rangle, \quad C_0 = \langle 1, z^8 \rangle$$

in $R = \mathbb{C}[z]/(z^{17} - 1)$, with $\lambda = [z^{16}]$, without the large exterior rank certificate. The quotient representatives are

$$U_A = z^3 \langle 1, z^2, \dots, z^{12} \rangle, \quad U_B = \langle z, z^3, z^5, z^9, z^{10}, z^{11}, z^{12}, z^{13}, z^{14} \rangle, \quad U_C = \langle z, \dots, z^7 \rangle.$$

Suppose spaces A, B, C of dimensions 3, 4, 3 satisfy the second-normal conditions.

Put $u = z^2$ and write $A = z^3 \mathcal{A}(u)$ with $\mathcal{A} \subset \mathbb{C}[u]_{\leq 6}$ of dimension three. The odd part of C has the form $z \mathcal{D}(u)$ with $\mathcal{D} \subset \mathbb{C}[u]_{\leq 3}$. The even part is contained in $\langle z^2, z^4, z^6 \rangle$. The AC conditions say that the coefficients of u^3, u^4, u^5, u^6 in every product ad , $a \in \mathcal{A}, d \in \mathcal{D}$, vanish. The even part contributes zero to these conditions by parity.

Pair $\mathbb{C}[u]_{\leq 6}$ with itself by the coefficient of u^6 . The condition is $\mathcal{A} \perp \mathcal{D} \mathbb{C}[u]_{\leq 3}$. If $e = \dim \mathcal{D} > 0$, then

$$\dim(\mathcal{D} \mathbb{C}[u]_{\leq 3}) \geq e + 3.$$

For example, choose a basis of $\mathcal{D}$ with distinct leading degrees; multiplication by $1, u, u^2, u^3$ produces at least $e + 3$ distinct leading positions. Since $\dim \mathcal{A} = 3$ in a seven-dimensional perfect pairing, $e + 3 \leq 4$, so $e \leq 1$.

If $e = 0$, the three-dimensional C is the entire even subspace. The BC conditions with $A_0 = \langle 1, z \rangle$ give six independent conditions on the nine-dimensional U_B . Thus $\dim B \leq 3$, a contradiction.

If $e = 1$, choose $0 \neq d \in \mathbb{C}[u]_{\leq 3}$. The four forms imposing the central coefficient conditions are independent: multiplication by d on $\mathbb{C}[u]_{\leq 3}$ is injective and the coefficient pairing is perfect. Hence

$$\mathcal{A} = \mathcal{A}_d := \{a \in \mathbb{C}[u]_{\leq 6} : [u^3]ad = \dots = [u^6]ad = 0\}, \quad \dim \mathcal{A}_d = 3.$$

We claim the six-dimensional sum $z^3 \mathcal{A}_d + z^{11} \mathcal{A}_d$ is direct and disjoint from $P = \langle 1, z, z^8, z^9 \rangle$.

Indeed, suppose $z^3 a(z^2) + z^{11} b(z^2) \in P$ modulo $z^{17} - 1$. Comparison of the even and odd residues forces

$$b = b_{\text{low}} + \beta u^3, \quad \deg b_{\text{low}} \leq 2, \quad a = \alpha u^3 - u^4 b_{\text{low}}.$$

Since $b \in \mathcal{A}_d$ and $\deg(bd) \leq 6$, its central coefficients vanish only if $\deg(bd) \leq 2$. The polynomial ring is a domain, so $\beta = 0$. Now

$$ad = u^3(\alpha d - u b_{\text{low}} d).$$

The polynomial in parentheses has degree at most three. Its coefficients must all vanish, so $d(\alpha - u b_{\text{low}}) = 0$. Again using the domain property gives $\alpha = 0$ and $b_{\text{low}} = 0$, and then $a = b = 0$. This proves the claim.

Consequently

$$\dim (A_0 C_0 + A C_0) = 4 + 6 = 10.$$

The first and second normal conditions put the full B function space in its annihilator, of dimension $17 - 10 = 7$. This annihilator contains B_0 , which has dimension four. Its quotient therefore has dimension at most three, again contradicting $\dim B = 4$.

This proof covers every nonzero complex d . The 40 rational coefficient examples checked by `cyclic_third_order.py` are implementation controls, not a substitute for this universal argument.

B A grid control for the pair conditions

In the ordered normal character bases of Section 7, the following columns span spaces of dimensions $(3, 4, 3)$:

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad C_1 = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$B_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Direct rational substitution gives all eight families of pairwise conditions in (15). For every triple of columns the cubic moment is zero. The consistency suite checks both facts. This demonstrates why merely extending the cyclic injectivity claim to the grid would be incorrect; the additional quadratic and flattening argument is needed.

Sources

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